

SIMATS ENGINEERING ASSESSMENT TOOL1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5117

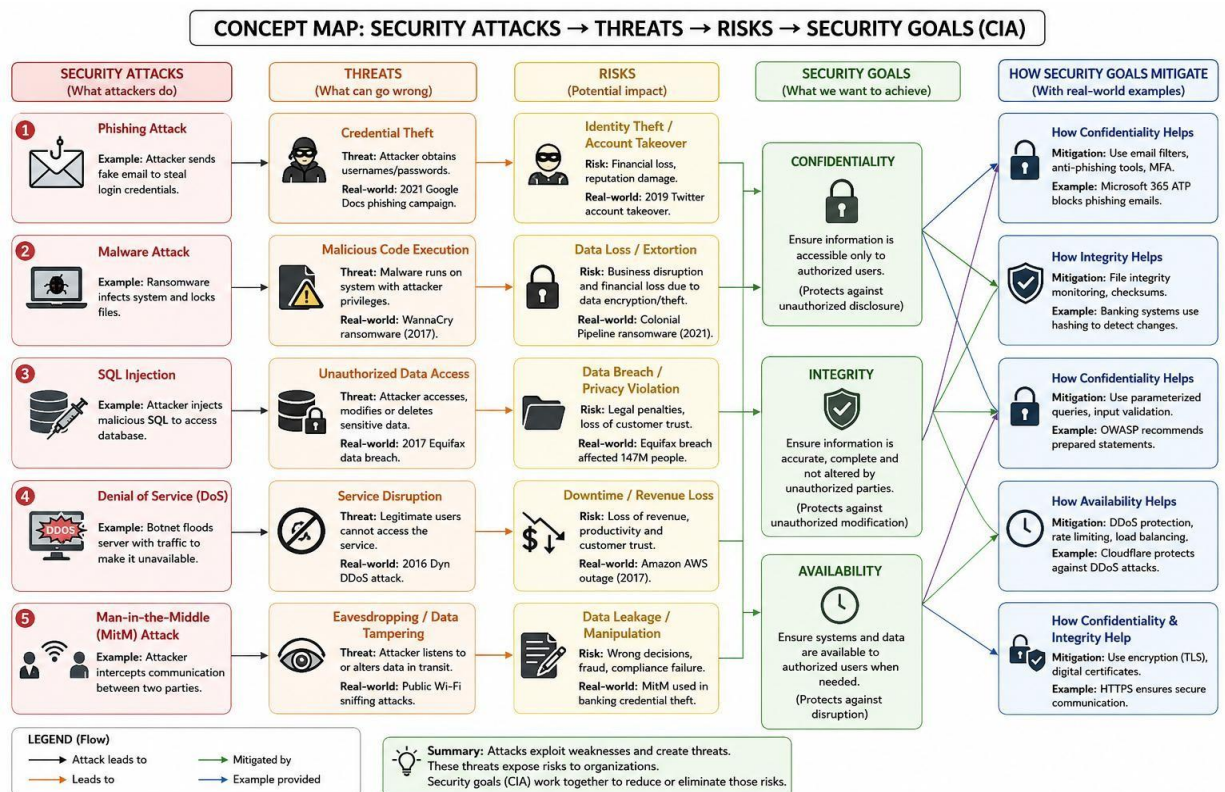
Name: VEERLA NAGA RAVINDRA
Reg.no: 192325067

CONCEPT MAPPING

Q1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:

- Show how each attack leads to specific threats and risks
- Map how security goals help mitigate them Provide one real-world example for each link

ANS:



Concept Map Overview

In cybersecurity, these core elements form a continuous lifecycle: An **Attack** (the action) materializes a **Threat** (the danger), which introduces a **Risk** (the business impact). We implement **Security Goals** (the CIA Triad) to break this chain and mitigate the risk.

Below is the conceptual mapping and detailed breakdown of how these elements link together, categorized by the core pillars of the CIA Triad.

Real-World Example:

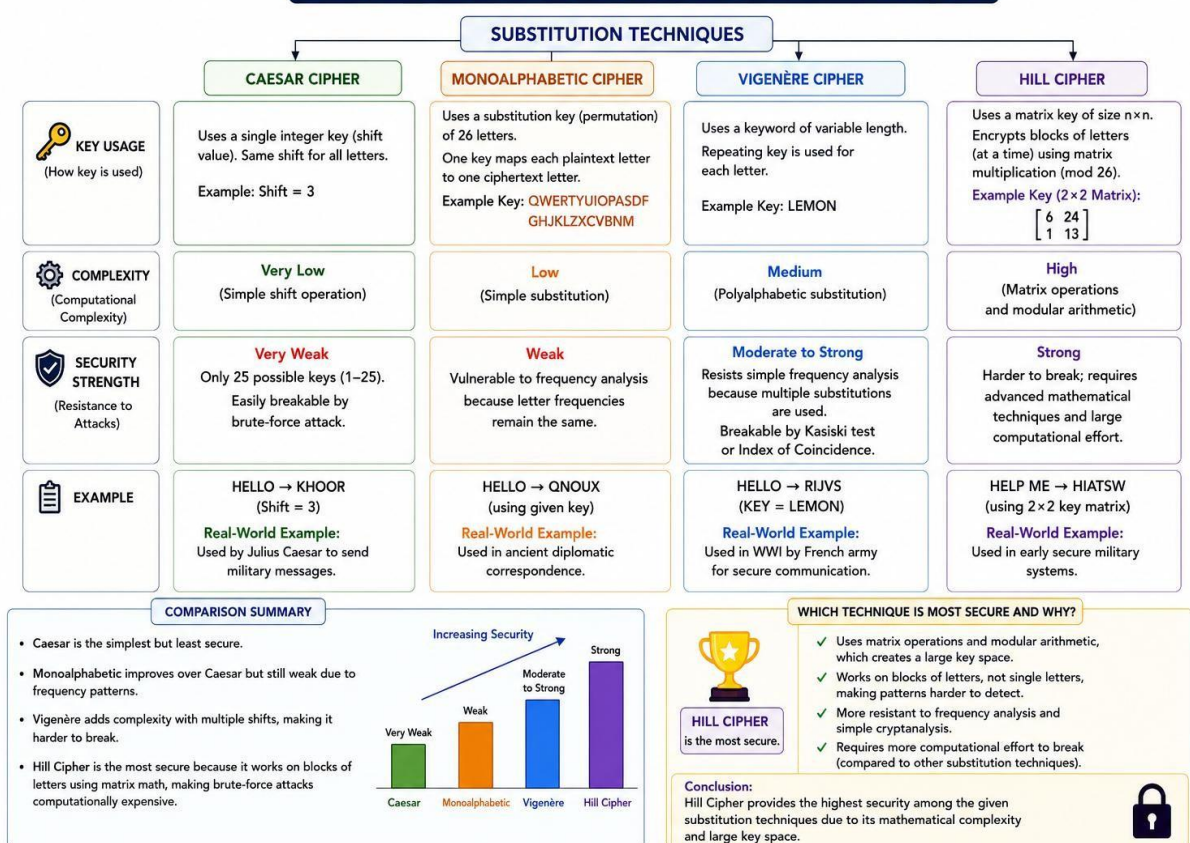
Link: A botnet floods an e-commerce website's servers with malicious traffic during a holiday sale (**Attack**), crashing the servers and causing a total service outage (**Threat**), which results in millions of dollars in lost hourly revenue (**Risk**).

Mitigation: Ensuring **Availability** through geographically distributed Content Delivery Networks (CDNs) and auto-scaling cloud infrastructure absorbs the malicious traffic, keeping the storefront online for legitimate customers.

Q2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

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2. CONCEPT MAP: COMPARING SUBSTITUTION TECHNIQUES



1. Concept Map: Comparison of Substitution Techniques

This mapping categorizes the four classical substitution ciphers based on their structural properties, establishing a clear progression from simple single-character shifts to complex mathematical transformations.

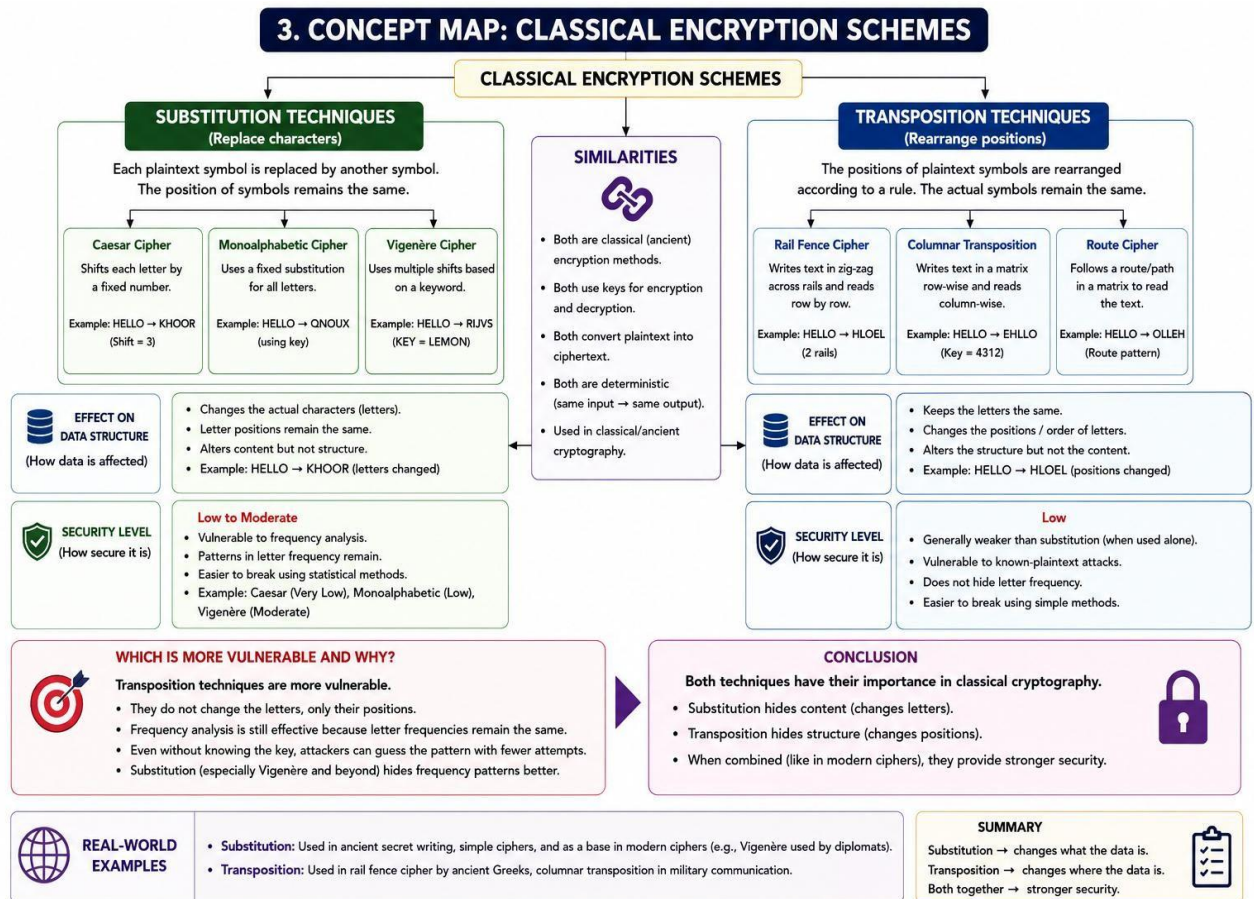
Cipher Technique	Key Usage	Complexity	Security Strength
Caesar Cipher	A single integer (1-25) representing a fixed alphabetical shift.	Very Low. Simple addition/subtraction mechanism.	Very Weak. Trivial to break via brute-force (only 25 possible keys).
Monoalphabetic	A randomly shuffled 26-letter alphabet (key space of $26!$).	Low. 1-to-1 character substitution mapping.	Weak. Defeated easily by linguistic frequency analysis and pattern recognition.

Cipher Technique	Key Usage	Complexity	Security Strength
Vigenère Cipher	A text keyword repeated to match the length of the plaintext.	Medium. Polyalphabetic; shifts change dynamically per character.	Moderate. Masks standard letter frequencies but vulnerable to Kasiski examination.
Hill Cipher	An invertible $n \times n$ matrix (where n is the block size of characters).	High. Polygraphic; requires matrix multiplication and modular arithmetic.	Strong (Relatively). Highly resistant to ciphertext-only frequency analysis.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

- a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why

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1. Concept Map: Substitution vs. Transposition (Similarities & Differences)

While both are classical, symmetric-key encryption methods that formed the foundation of early cryptography, they manipulate plaintext in fundamentally different ways.

Feature	Substitution Techniques	Transposition Techniques
Core Action	Replaces elements (e.g., 'A' becomes 'X').	Rearranges elements (e.g., 'ABC' becomes 'CAB').
Cryptographic Principle	Introduces Confusion (obscures the relationship between the plaintext and ciphertext).	Introduces Diffusion (spreads the plaintext structure across the ciphertext).
Character Identity	Altered (The original letters are hidden).	Preserved (The exact original letters are visible, just moved).
Similarities	1. Both use a secret key shared between sender and receiver.	

Feature	Substitution Techniques	Transposition Techniques
	2. Both operate on the plaintext alphabet. 3. Both are highly vulnerable to modern cryptanalysis when used in isolation.	

Substitution Techniques (e.g., Caesar, Monoalphabetic, Vigenère)

Data Structure Impact: Alters the value of the data at each specific index. The original sequence and spatial structure (like word lengths or repeating character positions) often remain entirely intact, but the characters themselves are transformed.

Security Level: Low to Moderate. While polyalphabetic substitution (like Vigenère) provides a moderate security level by masking single-letter frequencies, monoalphabetic substitution is extremely weak as it preserves the exact statistical footprint of the underlying language.

Transposition Techniques (e.g., Rail Fence, Columnar Transposition)

Data Structure Impact: Alters the index/location of the data while leaving the values untouched. The spatial relationships (words, consecutive letters) are shattered, scattering them across the data block.

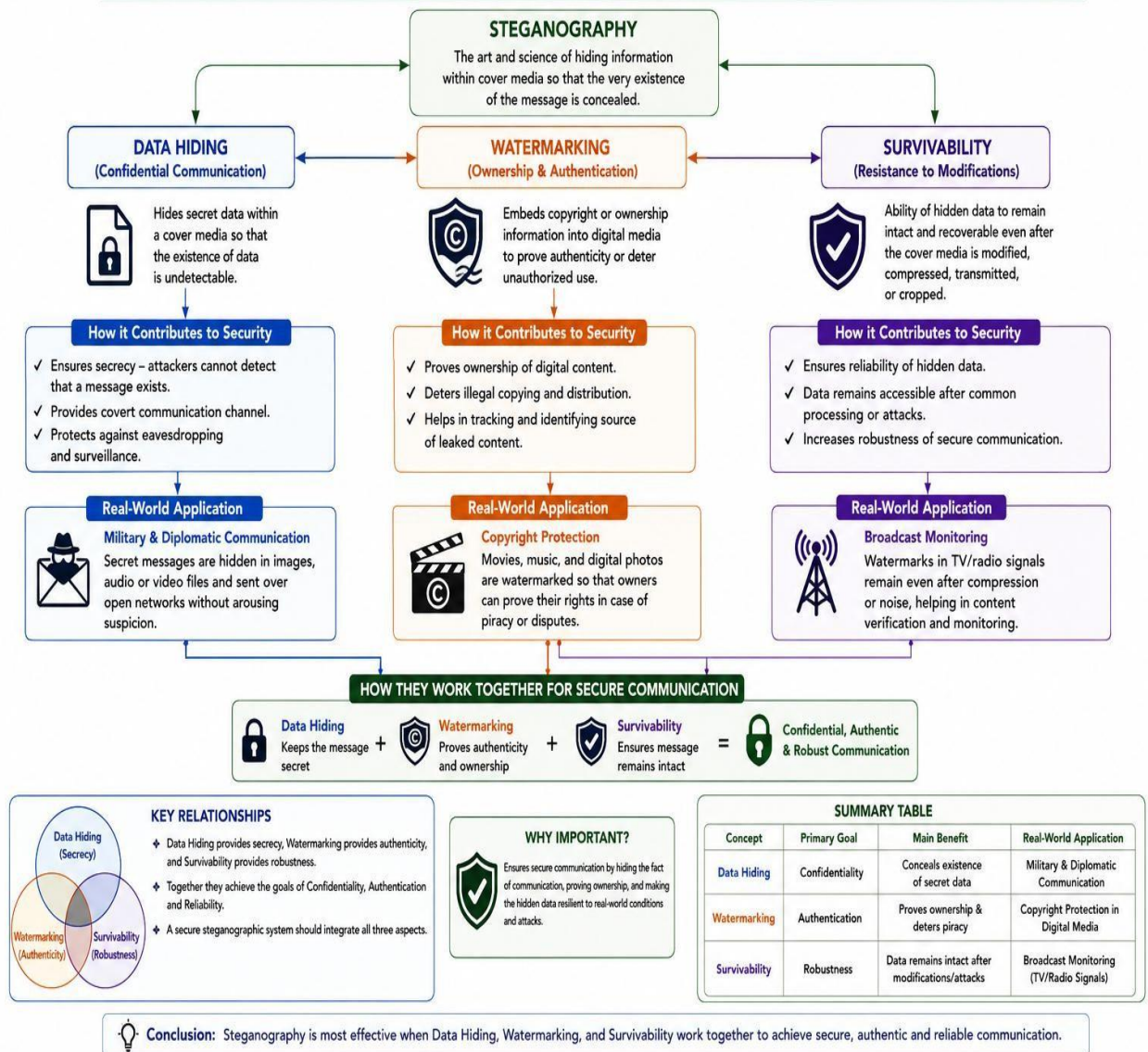
Security Level: Low. Because it only moves letters around, an attacker possesses the exact unencrypted characters of the message. Security relies entirely on the complexity of the routing/rearrangement algorithm.

Q4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

- Show relationships between these concepts
- Analyze how each contributes to secure communication
- Include one real-world application for each

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4. CONCEPT MAP: STEGANOGRAPHY – DATA HIDING, WATERMARKING & SURVIVABILITY



1. Concept Map: Relationships

Steganography is the overarching discipline of concealing information. Within this discipline, **Data Hiding**, **Watermarking**, and **Survivability** interact as follows:

Data Hiding is the fundamental mechanism. It is the process of embedding a payload (the secret) into a cover medium (image, audio, video).

Watermarking is a specialized application of Data Hiding. While standard data hiding uses a cover medium merely as a transport vehicle for a secret message, watermarking embeds data that is fundamentally tied to the cover medium itself (e.g., copyright info).

Survivability (Robustness) is the critical property/metric of the embedded data. It dictates whether the hidden data or watermark can endure alterations (like compression, cropping, or format conversion) without being destroyed.

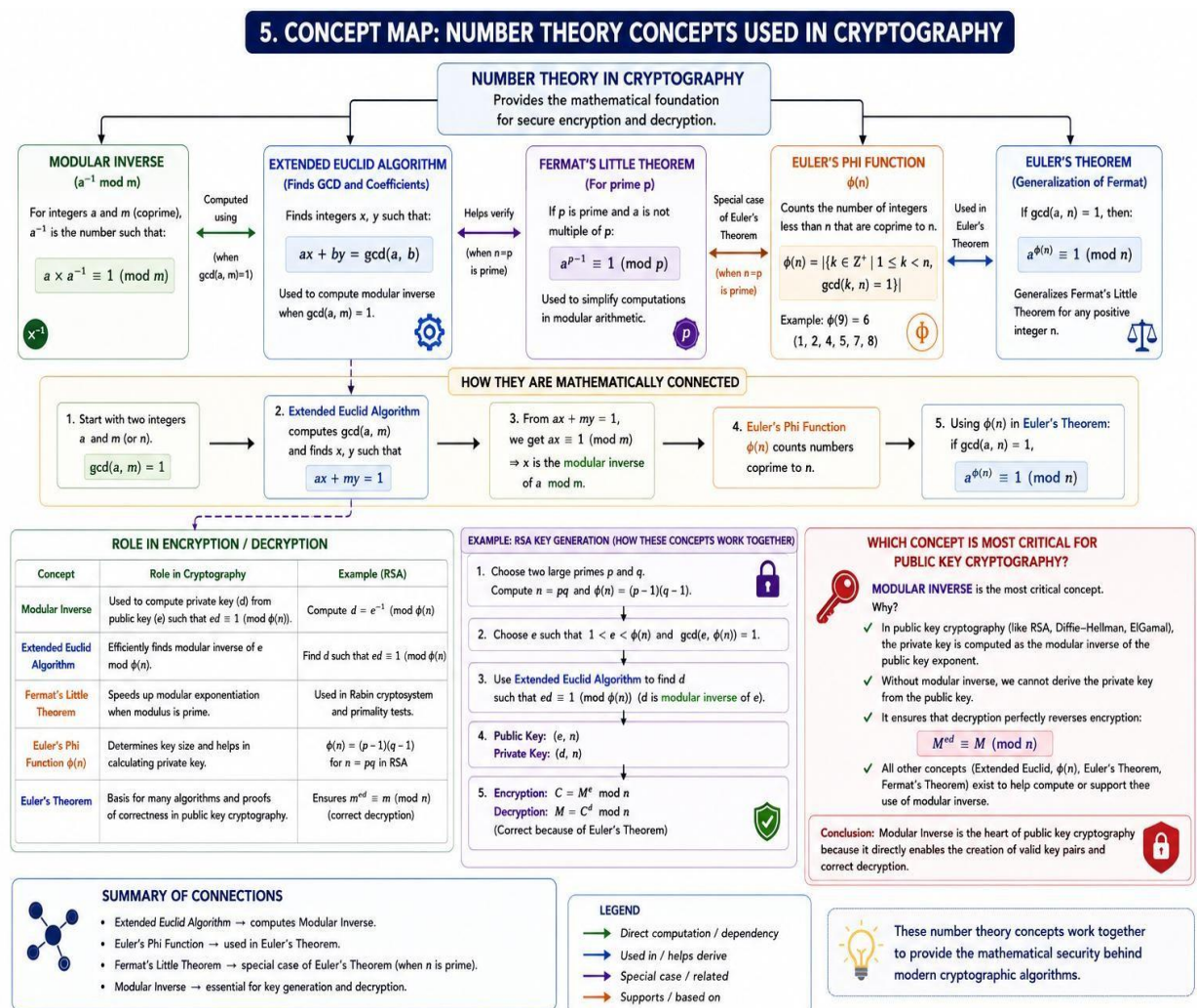
Visualizing the Link:

Data Hiding (The Method) $\rightarrow$ enables **Watermarking** (The Application) $\rightarrow$ which requires **Survivability** (The Defense).

Q5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

- Show how each concept is mathematically connected
- Map their role in encryption/decryption
- Analyze which concept is most critical for public key cryptography and justify

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1. Concept Map: Mathematical Connections

These five number theory concepts are deeply intertwined, forming the mathematical foundation for modern cryptography. Their relationships can be mapped as a progression from counting, to theorems, to algorithmic calculation:

Euler's Phi Function, $\phi(n)$: This is the starting point. It counts the number of positive integers up to n that are relatively prime (coprime) to n .

Euler's Theorem: This theorem relies directly on Euler's Phi Function. It establishes that if a and n are coprime, then:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

Fermat's Little Theorem (FLT): FLT is simply a **special case of Euler's Theorem** where the modulus is a prime number p . Since all numbers less than a prime p are coprime to it, $\phi(p) = p - 1$. Substituting this into Euler's Theorem yields FLT:

$$a^{p-1} \equiv 1 \pmod{p}$$

Modular Inverse: Both Euler's Theorem and FLT prove that modular inverses exist. An inverse of a modulo m is a number x such that $a \cdot x \equiv 1 \pmod{m}$. Using Euler's Theorem, the inverse can be expressed as $a^{\phi(m)-1} \pmod{m}$.

Extended Euclidean Algorithm (EEA): While Euler's Theorem proves the existence of the Modular Inverse, computing it using exponents is slow. The **EEA is the practical algorithmic tool** used to efficiently calculate the Modular Inverse by finding coefficients x and y such that $ax + by = \gcd(a,b)$.

Justification:

Public Key Cryptography fundamentally relies on a "**trapdoor one-way function**"—a process that is easy to compute in one direction but practically impossible to reverse unless you possess a special piece of secret information.

Euler's Phi Function provides this exact trapdoor.

If you know the prime factors p and q , computing $\phi(n)$ is trivial:
 $\phi(n) = (p-1)(q-1)$.

Because the private decryption key d can only be calculated if you know $\phi(n)$, the extreme difficulty of reversing Euler's Phi Function without the prime factors is the literal mathematical barrier that prevents an attacker from deriving the private key from the public key. Without this asymmetric difficulty, the entire paradigm of openly sharing a public key while keeping the private key secure would collapse.

